LESSON PLAN

EMI/EMC (14)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	30
Total No. of Lesson Plans	46	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Vertical speed hold and heading hold operation, Mach hold and need for air data computers
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Vertical speed hold and heading hold operation, Mach hold and need for air data computers.
5. Specific Objectives	14.2.1.1 Trainee will be able to explain the functions of AFCS.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of AFCS panel indications and mode annunciations; Instructor supervision required for safety.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the AFCS?
(b) How does the autopilot maintain altitude?
(c) Why is air data required for Mach hold?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Vertical Speed Hold Operation • Heading Hold Operation • AFCS Mode Annunciations	L & Ds	PPT, IB	15 min
(b)	• Mach Hold Operation Principles • Air Data Computer Functions • Integration of Flight Control Systems	L & Ds	PPT, PDF	20 min
(c)	• System Failures and Recovery • Pilot Intervention Procedures • Mode Transition Logic	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the role of the Air Data Computer in Mach hold.
(b) Describe the sequence of events during a heading hold engagement.
(c) What indications confirm successful vertical speed hold?

12. Home Assignment and Follow Up	Prepare a brief note on the ILS system, ILS coupling, VOR system, and VOR coupling for the next lesson.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Brief explanation of ILS system, ILS coupling and VOR system and VOR coupling.	Refer to the master syllabus reference list	LP 31

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min